Java1: Lesson 7 – Lab Project

1. Add **summary** at the top, **documentation** at the bottom and **comments** where necessary.
   - Recordings and interviews are an exception, but they should be included in your submission code.
2. **Cite** the source of your information.
3. Refrain from including anything **you don't fully understand**.

2

50 points

Run the code in IntelliJ using the **command-line mode** to display the result of 1 + 10

1. Add a new line of code to print out A.
2. Record the steps you take to run the code.
3. Refer to shorturl.at/fOVWX

```
public class test9 {
    public static void main(String[] args) {
        int result = 0;
        switch (args[1].charAt(0)) {
            case '+':
                result = Integer.parseInt(args[0]) + Integer.parseInt(args[2]);
                break;
        }
        String A = args[0] + ' ' + args[1] + ' ' + args[2] + " = " + result;
        //A =
    }
}
```

File(s) to submit	YouTube link

3

100 points

Utilize an array to produce an output similar to the sample output below.

1. Prompt the user to input 10 upper- and lower-case characters.
2. Implement your **own algorithm** to sort the characters in the array.
3. Insert comments (**//***) in your code to indicate where modifications were made to the textbook algorithm.

Modify the **sorting and **searching** algorithms from the textbook, ensuring **your implementation is different** from the original. Avoid using any built-in methods, such as 'sort()'.*

Sample output

```
10 upper or lowercase characters? j K L e w F v m p X
Ascending(1) or Descending(2) order? 1
=============
Your array : j K  L e w F v m p X
Ascending : e F j  K L m p V w X
-------------------------------------------------------------
10 upper or lowercase characters? j K L e w F v m p X
Ascending(1) or Descending(2) order? 2
=============
Your array : j K L e w F v m p X
Descending :  X w V p m L K j F e
-------------------------------------------------------------
10 upper or lowercase characters? 0
Ok done!
```

	File(s) to submit	J107_3.java J107_3.png
4	100 points Prompt the user to enter a single character (case sensitive), then compare it with the ten elements in an array. If the guessed character is present in the array, display the position(s) of the character in the array. **Modify the **sorting** and **searching** algorithms from the textbook, ensuring **your implementation is different** from the original. Avoid using any built-in methods, such as 'sort()'.*	
	Sample output	Guess one character (lowercase) in my array: a 1 2 3 4 5 6 7 8 9 0 My array: j a e l a w k v a k Good: A is in 2nd and 5th and 9th ---------- Guess one character (lowercase) in my array: k 1 2 3 4 5 6 7 8 9 0 My array: j a e l a w k v a k Good: k is in 10th ---------- Guess one character (lowercase) in my array: u 1 2 3 4 5 6 7 8 9 0 My array: j a e l a w k v a k Nope. ---------- Guess one character (lowercase) in my array: -1 Ok, done!
	File(s) to submit	J107_4.java J107_4.png